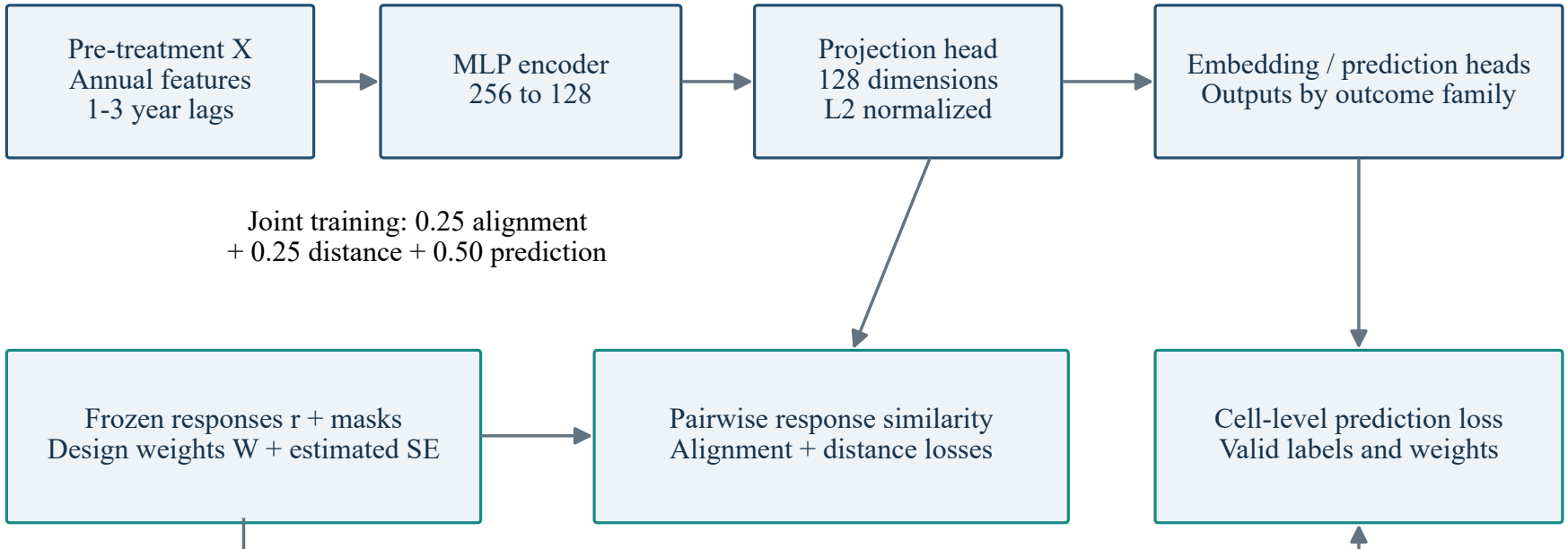
Pre-treatment X
Annual features
1-3 year lags
MLP encoder
256 to 128
Projection head
128 dimensions
L2 normalized
Embedding / prediction heads
Outputs by outcome family
Joint training: 0.25 alignment
+ 0.25 distance + 0.50 prediction
Frozen responses r + masks
Design weights W + estimated SE
Pairwise response similarity
Alignment + distance losses
Cell-level prediction loss
Valid labels and weights